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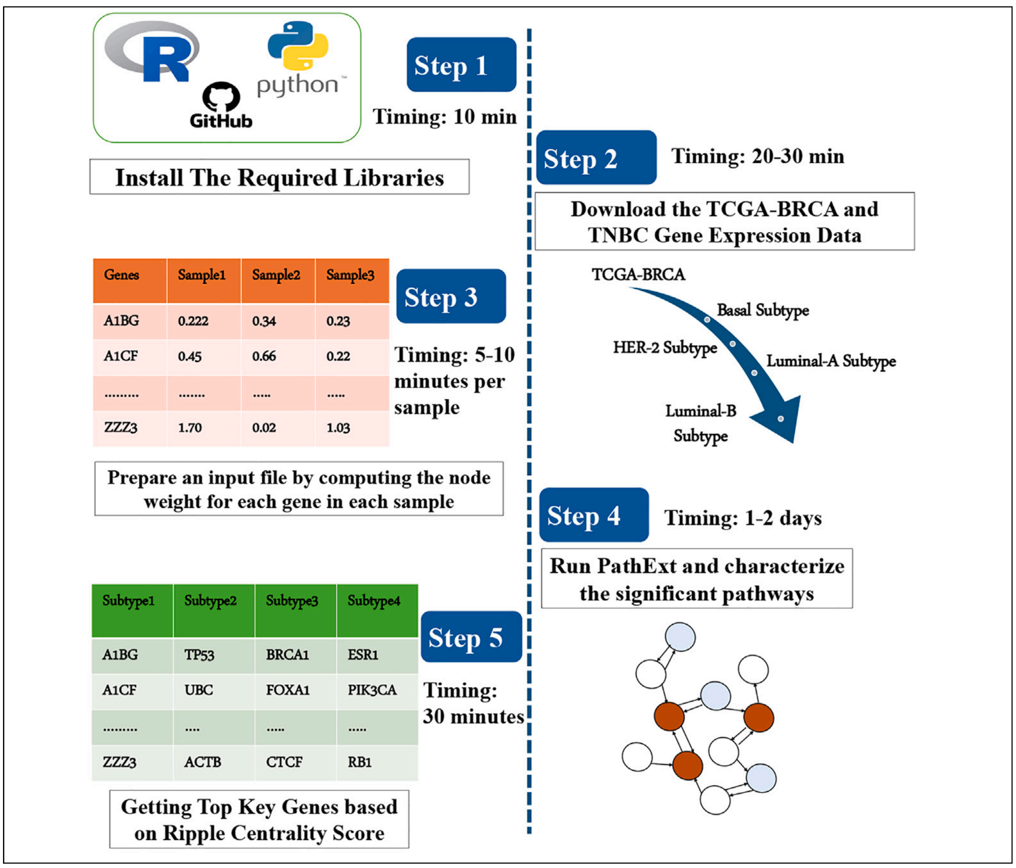
Protocol for identifying key genes using network-based approach as an alternative to differential expression analysis



In a variety of biological contexts, characterizing genes associated with disease etiology and mediating global transcriptomic change is a key initial step. Here, we present a protocol to identify such key genes using our tool "PathExt," a tool that implements a network-based approach. We describe steps for installing libraries, preparing input data and detailed procedures for running PathExt, and characterizing differential pathways and key genes based on ripple centrality scores.

Publisher's note: Undertaking any experimental protocol requires adherence to local institutional guidelines for laboratory safety and ethics.

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Highlights

Computation of BRCA subtype-specific central gene using network-based approach, PathExt

Steps for computing node weight for each sample using new statistical approach

Steps for executing PathExt to compute statistically significant pathways

Key gene identification for various analyses based on ripple centrality score

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Protocol for identifying key genes using network-based approach as an alternative to differential expression analysis

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SUMMARY

In a variety of biological contexts, characterizing genes associated with disease etiology and mediating global transcriptomic change is a key initial step. Here, we present a protocol to identify such key genes using our tool "PathExt," a tool that implements a network-based approach. We describe steps for installing libraries, preparing input data and detailed procedures for running PathExt, and characterizing differential pathways and key genes based on ripple centrality scores.

For complete details on the use and execution of this protocol, please refer to Agrawal et al.[1,2]

BEFORE YOU BEGIN

Given transcriptional data for cases and controls in a biological context, identification of DEGs is a conventional first step, with the potential to identify new biomarkers and targets. However, recent studies have reflected two major limitations with DEG based approach. First, in the DEG-centric approach, certain genes represent generic transcriptional responses and are not context specific, hence leading to false positives. Second, global transcriptional changes represent the terminal events and not necessarily the key regulators of those changes.[3,4]

Alteration in crucial mediators (such as transcription factors, kinases, and other regulatory proteins) within an intricate gene regulatory network results in modifications to the transcriptome. Finding these mediators, as opposed to the terminal events, is more likely to provide mechanistic insights. It has been demonstrated that network-based models enhance potential target identification in a variety of malignancies.[5]

The provided protocol discusses our network-based tool "PathExt" and its implementation to characterize key genes in Breast Cancer (BRCA) subtypes in step-by-step manner. PathExt provides an alternative to DEG based approach by characterizing statistically significant "differentially active paths". Sub-network composed of such paths capture the highly relevant genes and processes underlying the disease context, and the central genes in this sub-network represent the potential mediators of the global transcriptomic changes. Here we implemented this approach using transcriptome data, however, it is worth noting that this protocol can also be implemented using other data (epigenomic, proteomic, etc.). Furthermore, this approach can be adopted even if there is no control available.[6]

Computational requirement

PathExt is compatible with Unix/Linux operating systems. Users need to install specific packages and dependencies to run it which include conda (for easy installation of other packages), Python version 3.7 or lower; R package (preferably 3.6 or lower); pandas; numpy, networkx 1.11; and statsmodel. As PathExt utilizes a large network of ∼20,000 nodes and ∼300,000 edges, the computer should have at least 16GB RAM and 4 CPUs per core.

Input files

Timing: 30 min

1. Prepare input files for PathExt.
   a. Pre-processed normalized gene expression data of breast cancer samples and control from TCGA-BRCA cohort and TNBC transcriptome data.
   b. Knowledge-based network comprising experimentally validated protein-protein and regulatory interactions, downloaded from study.

Node weight computation and PathExt implementation

Timing: 1–2 days

2. The transcriptomic data is projected onto the knowledge-based network and the node-weight for each gene is computed using the R package. The node-weight can be computed in a sample-specific fashion, or for a cohort of samples, and it represents relative expression of a gene in a sample or cohort relative to control.
3. The node-weighted network is provided as input to PathExt (can be downloaded https://github.com/agrawalpiyush-srm/PathExt-BRCA), which first identifies the subnetwork called TopNet, composed of differentially active paths.

Characterizing key central genes from TopNet

Timing: 1 h

4. Compute Ripple centrality of each gene in TopNet.
5. Select top genes based on ripple centrality (high to low) per sample.
6. Prioritize BRCA subtype-specific key genes based on the frequency with which a gene is selected among the central TopNet genes across the subtype-specific samples.

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
Breast cancer Bulk RNA-seq	TCGA[7]	https://portal.gdc.cancer.gov/
TNBC Microarray Dataset	Horak et al.[8]	GEO: GSE41998
TNBC RNA-seq Dataset	Chen et al.[9]	GEO: GSE163882
Protein-Protein Interaction Network	Shukla et al.[10]	https://elifesciences.org/articles/25624
Software and algorithms		
PathExt	Sambaturu et al.[6] and Agrawal et al.[1]	https://github.com/agrawalpiyush-srm/PathExt-BRCA
Pandas	PyData	https://pandas.pydata.org/
Numpy	NumPy	https://numpy.org/
R	CRAN	https://cran.r-project.org/

(*Continued on next page*)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Python 3.6.9	Python	https://www.python.org/downloads/release/python-369/
Statsmodel	Statsmodels	https://www.statsmodels.org/stable/index.html
Networkx 1.11	Python	https://networkx.org/documentation/networkx-1.11/
DESeq2	Love et al.[11]	https://bioconductor.org/packages/release/bioc/html/DESeq2.html
biomaRt	Durinck et al.[12]	https://bioconductor.org/packages/release/bioc/html/biomaRt.html

STEP-BY-STEP METHOD DETAILS

Input file preparation

Timing: 20–30 min

This step is required to create an input file needed to compute node weights. Following are the key steps.

1. Download the raw count matrix of breast cancer gene expression data from Genomic Data Commons data portal in csv file format.
2. Map the ENSEMBLE id with the Gene Symbol using the following R command. This will take the read count matrix file as an input and provide the output file where ENSEMBL IDs have been converted to Gene Symbol format.

```
>Rscript ensembel_genesymbol.r
```

Note: We do the above mapping as PPI network has gene information in the form of symbols. For conversion, the biomaRt package in R is used.

3. Normalize the raw read count matrix in the log2(tpm + 1) format using the following code.

```
>Rscript compute_logtpm.r
```

Note: This code will take the raw count matrix file with Gene Symbol as an input (output file of above command), implement DESeq2 package and will provide output file with log(tpm+1) values for each gene.

4. After doing log transformation, quantile normalize the case and control data together using the following R commands.

```
library (limma)

data <- read.csv (``log_TPM_normalized_data.csv'', header = FALSE)

data_mat <- data.matrix(data)

norm_data <- normalizeQuantiles (data_mat)

write.csv (norm_data, file = ``Quantile_Norm_data'')
```

5. Once the data is quantile normalized, separate the data in subtype-specific manner and compute the node weights in subtype manner as mentioned in the next section.

Node-weight computation

Timing: 5–10 min per sample

This step is required to prepare the input file for PathExt to characterize the TopNets in a sample. The key step here is to compute Expected Log Fold Change by regressing the fold change against the control gene expression using 'Loess fitting' and then using the ratio of Observed over Expected Fold Change as the node(gene) weight. The subtype-specific node weight matrix was created for PathExt in the following manner. To compute node weight of each gene in a sample.

6. Download R from here and install it into the system by running the following commands.

```
> tar -xzvf R-4.3.1.tar.gz
> cd R-4.3.1
> ./configure –prefix=/usr/local
> make
> sudo make install
```

Note: Installation varies as per operating system. Here, we used the Unix operating system and installed the R by running the following commands.

7. When both the case and control is present.
   a. Prepare an input file "act_input.csv" with 3 columns (i) gene Id (ii) expression of the gene in foreground or case sample (i.e., case); (iii) average expression of the gene in control samples. Run the following command:

```
> data <- read.csv("act_input.csv", header = TRUE)
>head (data)
gene, control, case
A1BG, 0.2274, 0.081
A1BG-AS1, 1.2395, 1.4597
A1CF, 0.0229, 0.0521
A2M, 6.3369, 5.3741
A2M-AS1, 0.6236, 0.5077
A2ML1, 5.6892, 5.4869
```

Note: This file is required to compute the node weight for each sample.

   b. Once the input file is ready, user needs to run the "Node_weight_computation.r" function written in R to compute the node weight as follows.

```
> Rscript Node_weight_computation.r act_input.csv act_output.csv
```

   c. Once the above command is run, we will have an output file "act_output.csv". The output file will consist of gene names in sorted order along with the corresponding node weight, which will look as follows.

```
> output <- read.csv("act_output.csv", header = TRUE)
>head (output)
gene, node_weight
ABHD16B, 0.0019
ACTG1P4, 0.0019
ADAM1A, 0.0019
AHCYP8, 0.0019
AKAP17A, 0.0019
```

8. When the control is not present.
   a. In case, the control is not available, we take normalized gene expression directly as the node weight or another option can be computing the *Z* score of the gene in the sample computed relative to all the samples in the matrix.
9. Remove genes that are not present in knowledge-based network. This is an optional step, as genes that are common in node weight matrix files and protein-protein interaction files will only be used to create interaction networks.

   Note: The above code will compute the node weights for the Activated TopNet i.e., paths differentially active in case (cancer patient) with respect to control. To compute the node weights for the Repressed TopNet, you need to compute the fold change in reverse order i.e. fc = (control/case) and run the remaining code as above.

PathExt implementation to compute sample-specific TopNet

Timing: 1–2 days

This step is the main step of the protocol where PathExt characterizes differentially perturbed paths in BRCA and TNBC samples for each subtype. Subsequently, we characterize sub-networks (TopNets) to obtain the key genes. This step can be further categorized into steps as follows.

10. PathExt requires several software, dependencies and packages to be installed.
    a. Download the 'conda' installer, as it helps in installing software and dependencies easily and can be downloaded here.
    b. Once the installer is downloaded, install it by running following command on the terminal:

```
bash Miniconda3-latest-MacOSX-x86_64.sh -b -p $HOME/miniconda
```

    c. Next, create an environment with Python version 3.7. or lower.

```
conda create -n myenv python = 3.6.9
```

    d. Next, activate the environment 'myenv' and installed the required libraries and software.

```
conda activate myenv
conda install pandas=0.25.3 numpy networkx=1.11 statsmodels
```

11. TopNet computation using PathExt in BRCA and TNBC subtypes.
    a. When the node weight is computed using case and control both.

i. Download the PathExt folder with the python codes from https://github.com/agrawalpiyush-srm/PathExt-BRCA.
ii. Create the folders "test_data" and "results" for the ease of working using the command.

```
> mkdir PathExt/test_data

> mkdir PathExt/test_data/results
```

iii. Copy the input file of sample specific node weights and knowledge-based network file into the 'test_data' folder.

```
> cp input_file PathExt/test_data/

> cp human_PPI.txt PathExt/test_data/
```

Note: Input_file will be different for the BRCA and TNBC subtypes.

iv. Build a network with PPI and node weight matrix as inputs and monitor the top 'X' percentile shortest paths in the network. To further assess the significance of each top shortest path, we permute the input node weights to produce a negative control.

```
> cd PathExt/

>mkdir test_data/results/temp

> python node_weight_matrix_colname.Pijs.py test_data/input_data Sample1 test_data/hu-
man_PPIN.txt 0.005 2 200 test_data/results/Activated_Response test_data/results/temp/Pij
```

Note: Users can encounter an error at this point of step if all the required libraries are not installed or if the name of the Sample in the code doesn't match with that of the input file or the input file is not tab separated. Please look at the troubleshooting section; problems 1, 2, and 3.

v. Significance of a path's length is estimated nominally from the distribution of path lengths (between the same two nodes) in the negative randomized control.

```
> python fdr_rand_pijs_boxcox.py test_data/results/temp test_data/results/Pij_
zscores.txt
```

vi. Given the nominal p-value for each of the top shortest paths, we adjust it for multiple testing.

```
> rm -f test_data/results/temp

> python benjamini_hochberg_boxcox.py test_data/results/Pij_zscroes.txt 0.05 test_data/
results/Pij_zscroes_fdr.txt
```

vii. Significant paths (after multiple testing correction) comprise the TopNet.

```
> python extract_fdr_network.py test_data/results/Activated_Response test_data/results/
Pij_zscores_fdr.txt 0.05 test_data/results/ Activated_Response_TopNet.txt
```

b. When node weight is computed using only case (in the absence of control).
   i. When the control is absent, we capture the highly active processes in the condition of interest. To get the same, run the following command.

```
> python get_highest_activity_TopNet.py test_data/input_sample Sample1 test_data/human_-
PPIN.txt 0.005 2 test_data/results/HA_base_network.txt test_data/results/HA_TopNet.txt
```

12. Compute Ripple Centrality Score.
    a. Post TopNet generation, ripple centrality score for each node is computed by running the following command:

```
> python calc_ripple_centrality.py test_data/results/ Activated_Response_TopNet.txt
test_data/results/ Activated_epicenter
```

13. Extract Central Genes based on Ripple Centrality Score.
    a. Arrange the list of genes based on decreasing order of the ripple centrality score.
    b. Select the top 'X' (user selected parameter) genes; in our applications, we selected the top 100 central genes.
    c. Get top 'X' key genes in each sample, and further compute the frequency of each key gene across case samples.
    d. Lastly, select the top 'X' (user selected parameter) most frequent central genes; in our applications, we selected top 200 genes.
    e. Perform analysis for each of the four BRCA and TNBC subtypes to get subtype-specific central genes.

EXPECTED OUTCOMES

This protocol allows the user to characterize the key transcriptional regulator mediating the observed transcriptomic changes, often missed by the DEG-centric approach. These subtype-specific characterizations of key genes will allow the user to identify subtype-specific enriched biological pathways elucidating the underlying transcriptional heterogeneity and mechanisms in various BRCA subtypes. Key genes associated with a TNBC subtype further helps in elucidating the underlying mechanism associated with resistance to given treatment.

Importantly, this pipeline can be applied to any cancer type or in any diseased condition where two conditions (case and control) are present by integrating knowledge-based network with omics data (transcriptome, epigenome, etc.) In the event that an appropriate control is not available, the approach elucidates the most active paths in a given condition. The protocol has an extra added advantage over the DEGs as DEG-centric approaches' performance depends on the availability of large cohort of cases and controls.

Overall, the current protocol can outperform conventional DEG based approaches in identifying key regulators associated with heterogeneity in groups, new biomarkers or targets for better diagnosis and treatment.

LIMITATIONS

One of the limitations of PathExt pipeline is its dependency on knowledge-based protein-protein interaction network. Secondly, PathExt depends on specific packages and its version. For example, PathExt can be implemented only with python version 3.7 or lower. Third limitation is the system requirement and estimated time per sample to implement PathExt. PathExt creates a huge network of ~20,000 nodes and ~300,000 edges, and for this heavy computation, the computer should have at least 16 GB RAM and 4 CPUs per core which takes around 10 h to run. The time may go up to 24 h to complete depending upon the complexity of the input data and the generated network.

TROUBLESHOOTING

Problem 1

While running the code "node_weight_matrix_colname_Pijs.py", you encountered the following error.

```
Traceback (most recent call last):

File "/Users/piyushagrawal/Downloads/HPV/HPVPos_HPVNeg/PathExt_res/threshold_result/PathExt/
node_weight_matrix_colname_Pijs.py", line 91, in ⟨module⟩

Pij, node_weight = combine_data_get_sp_paths_costs_basenw(G_ref, node_weight, base_nw_
fname, []) # Give an empty list as the paths argument

File "/Users/piyushagrawal/Downloads/HPV/HPVPos_HPVNeg/PathExt_res/threshold_result/
PathExt/node_weight_matrix_colname_Pijs.py", line 42, in combine_data _get_sp_paths_
costs_basenw

Pij = net_fun.get_all_sp_paths_costs(G_base)

Fispath.costs.r-po.pataPrane.fo.dc.ountsppers.dics/th:gt_path.cat6, saths.dkcti0) for i
in spaths dnet.Keyst) for ghin spaths-

dictlil.keys() if i!=j}, orient='index')
```

Potential solution

This kind of error generally occurs when the environment where all the software is installed is not activated. User needs to activate the environment. For example, by typing the following command.

```
`conda activate myenv`
```

Also, make sure that node weight of any gene should not be '0'. User can simply substitute '0' with small value such as 0.0001 as this substitution does not change the result notably.

Problem 2

While running the command to build a network with PPI and node weight matrix inputs and monitor the top 'X' percentile shortest paths, we encounter following error.

```
File "/Users/piyushagrawal/miniconda3/lib/python3.12/site-packages/pandas/core/indexes/base.
py", line 3812, in get_100 raise KeyError (key) from err

KeyError: 'Sample1'
```

Potential solution

This kind of error generally occurs when the name of the 'Sample Id column' doesn't match with the correct name. Check the header of the input file and make sure that the name in the header matches with the name in the code for which TopNets are getting generated.

Problem 3

Sometimes, during the PathExt run, we encountered following error.

```
(myenv) piyushagrawal@Piyushs-MacBook-Pro PathExt % python node_weight_matrix_colname_-
Pijs.py test_data/input_data TCGA-BA-4077-01B test_data/human_PPIN.tx

t 0.5 2 200 test_data/results/Activated_response test_data/results/temp/Pij

Traceback (most recent call last):

File "/Users/piyushagrawal/miniconda3/envs/myenv/lib/python3.6/site-packages/pandas/
core/indexes/base.py", line 2897, in get_100 return self._engine.get_loc (key)
```

```
File "pandas/_libs/index.pyx", line 107, in pandas._libs.index.IndexEngine.get_10c File "pandas/_libs/index.pyx", line 131, in pandas._libs.index.IndexEngine.get_10c

File "pandas/_libs/hashtable_class_helper.pxi", line 1607, in pandas._libs.hashtable.-PyObjectHashTable.get_item File "pandas/_libs/hashtable_class_helper.pxi", line 1614, in pandas._libs.hashtable.PyObjectHashTable.get_item

KeyError: 'TCGA-BA-4077-01B'
```

Potential solution

This error comes when the input file is not 'tab separated'. PathExt takes input file in 'tsv' file format.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Piyush Agrawal (piyusha@srmist.edu.in).

Technical contact

For any technical issues contact Piyush Agrawal (piyusha@srmist.edu.in).

Materials availability

This study did not generate new unique reagents.

Data and code availability

This paper analyzes existing, publicly available data. These accession numbers for the datasets are listed in the key resources table.

Archived version of the code is available at Zenodo; (https://doi.org/10.5281/zenodo.13943929) and URL (https://doi.org/10.5281/zenodo.13943929).

This paper does not report the original code.

Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

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AUTHOR CONTRIBUTIONS

Conception/design, P.A. and S.H.; data acquisition, P.A.; data analysis/interpretation, P.A. and S.H.; writing – original draft, P.A.; writing – review and editing, S.H.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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